

# Patrick Wagstrom

Data Science, Engineering, and Artificial Intelligence Leader

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Executive leader of cutting edge data science, machine learning, and software engineering organizations. I accelerate value from data by combining best practices from machine learning, risk management, cloud deployment, devops, and human-computer interaction. Hands-on technical leader that builds cohesive distributed teams by bringing together great people from diverse backgrounds while creating a shared sense of vision and ownership.

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## Professional Experience

### **Brightcove, Boston, MA and Hartford, CT**

#### **Chief Data Officer**

May 2021 - December 2022

- Responsible for end-to-end analytics collection and processing system for tens of thousands of events a second, hundreds of millions of videos, billions of monthly views, and trillions of rows of data in our data warehouse.
- Built and managed an organization of 34 FTE and 12 contractors in engineering, data science, and product management responsible for data platforms, analytics and insights, data and model governance, and customer facing data products.
- Executive sponsor and owner of the \$12.3mm acquisition of Wicket Labs - Brightcove's biggest acquisition in 8 years that brought critical media analytics skills and focus to Brightcove. Brought the product to more than 50x as many customers in under 1 year.
- Ownership multi-million dollar annual relationship with Google for Google Cloud Platform.
- Led team and built out models of video classification from text transcripts, audio, and video processing - including sourcing of training data, development of algorithms, and creation of model execution platform.
- Rearchitected and led migration of Brightcove's real-time analytics pipeline from Google Dataflow to Google Pub/Sub, resulting in a savings of \$300k annually.

### **Verizon, Basking Ridge, NJ and Hartford, CT**

#### **Director of Emerging Technology**

September 2019-April 2021

- Built a team 15 FTEs and additional 51 vendor contractors with a \$15mm annual budget from the ground up - including attracting internal and external talent and negotiating for budget allocations.
- Defined and architect overall enterprise strategy for reproducible machine learning and MLOps including data management, model training, model serving, model refit, and model risk management.
- Led the advanced data warehouse and data pipeline architecture functions - migrating from a number of on prem ad-hoc solutions (i.e. Teradata and Hadoop) to a streaming cloud based solution around Google BigQuery.
- Led proof-of-concept solutions around blockchain identity and asset management, augmented and virtual reality for Verizon retail and 5G UWB Home Internet, cross-device biometrics, and more.
- Sourced data, built, and delivered critical models and dashboards for Verizon's COVID shutdown response, including influence on which retail locations should remain open.
- Conceived of and built models for cell site energy efficiency resulting in a \$5mm annual savings and

conversion into a digital twin for future simulation.

- Led team that developed a holistic model for connecting online brand discussion to customer profiles to support customer servicing - including customer and agent action prediction models and patent pending bot detection model.
- Worked with legal and policy experts for the creation of numerous Verizon policies, including facial recognition, energy efficiency, and data privacy.

## **Capital One, New York, NY and McLean, VA**

### **Senior Director of Data Science for Machine Learning Platforms**

January 2019-September 2019

### **Director of Data Science for Machine Intelligence**

November 2016-January 2019

- Architect and overall lead of 37 people building the Capital One Card Machine Learning Platform - a platform for scalable and personalized real-time reinforcement learning based machine learning models on top of Kubernetes, gRPC, and AWS inside a regulated industry.
- Led a team that built and deployed multi-modal customer intelligence models working on both structured and unstructured (text, voice) data resulting in \$27mm annual savings.
- Hired, managed, and developed a distributed team of 18 data scientists.
- Built out a \$1.2mm academic research partnership with universities to explore implications of fairness in artificial intelligence.
- Conducted more than 300 hiring interviews of data scientists, data engineers, and product managers to help grow Card Machine Learning from 33 to 168 people and Capital One's New York Card team from 8 to nearly 200.

## **IBM Watson, Littleton, MA**

### **Research Staff Member/Technical Lead**

January 2015-November 2016

- Led a globally distributed team that built the IBM Watson Conversation service - an innovative service to create rich interactions using natural language processing, entity recognition, and scripted dialog.
- Global team lead for Watson Developer Cloud Tooling. Created cutting edge applications for creating, training, and maintaining cognitive and machine learning solutions including Watson Engagement Advisor and IBM Watson Natural Language Classifier.
- Conducted research on optimal methods to build and configure cognitive and machine learning systems resulting in a 95% reduction in human time required to train these systems.
- As a subject matter expert and research, traveled around the world to conduct research with and work side-by-side with our customers.
- Led the charge evolve internal development standards to use a best-of-breed modern web application architecture based on Angular, Node.js, GitHub, Slack and more.

## **IBM Watson - Watson Life, Yorktown Heights, NY**

### **Research Staff Member**

January 2014-January 2015

- Leadership team member responsible for evaluating promising consumer applications of cognitive computing and guiding teams in lean startup and IBM design thinking processes to explore ideas.
- Organized and led globally distributed teams on how to contextualize solutions to cognitive computing challenges.
- Project lead for the IBM Food Truck at SXSW which demonstrated cognitive computing to more

than 4,000 people and resulted in more than 1 billion media impressions.

- Planned and executed international workshops on design thinking methodologies.

## **IBM TJ Watson Research Center, Yorktown Heights, NY**

### **Research Staff Member**

August 2009-January 2014

- Team lead for the "Millennial Enterprise" aspect of IBM's 2013 Global Technology Outlook, which described the need for user data in building out AI/ML and led to me being a founding research scientist for IBM Watson Group.
- Analytics lead for JazzHub, IBM's cloud software development strategy. Designed analytics strategy, implemented metrics driven development, introduced A/B testing, and developed analytics dashboards.
- Technical and strategy contributor to IBM's research and acquisition strategy for software development tools and processes.
- Developed and designed GitMiner - an open source project used by 15 universities to perform graph analysis on large scale software engineering databases such as GitHub and BitBucket.
- Led a research team to evaluate productivity of new users and small teams using IBM's enterprise software engineering and product development environments.
- Published papers on topics around distributed collaboration, technical debt in software, and flow of ideas in software engineering communities.
- Mentored three Ph.D. student interns on projects related to collaboration in software engineering

## **Carnegie Mellon University, Pittsburgh, PA**

### **Graduate Research Assistant**

August 2003-July 2009

- Designed and developed CVSMiner an open source tool to perform social network and technical analysis of software engineering ecosystems such as GNOME and Eclipse.
- Worked with members of the GNOME Foundation and Eclipse Foundation to evaluate and improve the relationships between non-profit foundations that manage open source ecosystems and commercial firms.
- Utilized a variety of qualitative and quantitative research methods: stakeholder interviews, message analysis, natural language processing, data mining, machine learning, and social network analysis to generate insight into largely ad hoc software development processes.

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## **Education**

### **Ph.D. in Engineering and Public Policy and Computation, Organizations, and Society, May 2009**

Carnegie Mellon University, Pittsburgh, PA

Thesis: "Vertical Interaction in Open Software Engineering Communities". Advisors: Dr. James Herbsleb and Dr. Kathleen Carley.

### **MS in Computation, Organizations, and Society, May 2007**

Carnegie Mellon University, Pittsburgh, PA

### **MS in Computer Science, August 2003**

Illinois Institute of Technology, Chicago, IL

Thesis: "Scarlet: A Framework for Context Aware Computing". Advisor: Dr. Xian-He Sun.

### **BS in Computer Science / BS in Computer Engineering / BS in Electrical Engineering, May 2002**

Illinois Institute of Technology, Chicago, IL